

Peer review

Team Ordinary Most Squares

- **Describe the goal of the project.**

The goal of the project is to predict academic performance based on different demographics and institutions.

- **Describe the data set used in the project. What are the observations in the data? What is the source of the data? How were the data originally collected?**

The data set contains 1000 observations, each observation being a secondary/post-secondary student. Each observations come with 16 variables about study hours, sleep hours etc.. No source of data or collecting methods mentioned.

- **Consider the exploratory data analysis (EDA). Describe one aspect of the EDA that is effective in helping you understand the data. Provide constructive feedback on how the team might improve the EDA.**

What is the difference between predicted exam scores and the exam score? Why are some EDA using the predicted while some use the true exam score?

You might have to read through your eda write ups, there are a few typos, such as figure numbers, inconsistency in narration perspective as well as wrong interpretations of graphs (right/left-skewed).

All of your plots are explanatory and display interesting relations between your response and prediction variables. You include a wide variety of plot types, which is very appealing to the reader, especially figures 5 and ,6 where you incorporate colors.

- **Describe the statistical methods, analysis approach, and discussion of model assumptions, diagnostics, model fit.**

You used multiple linear regression to model your response variable. There were a few potentially related predictor variables initially chosen, as well as some possible interaction effects between them, and you then selected the best model evaluating AIC, BIC and R^2 . Then, with the newly chosen predictors, you ran linearity, normality and homoscedasticity test. Then, a final model was chosen.

- **Provide constructive feedback on how the team might improve their analysis. Make sure your feedback includes at least one comment on the**

statistical modeling aspect of the project, but also feel free to comment on aspects beyond the modeling.

I like how you checked the linear model assumptions for your model.

It would be useful if we saw some of the process of finding the best model or testing the usefulness of the model. Moreover, you could display the r^2 and the RMSE of the final chosen model for the audience to grasp the true predictability. Moreover, I am just wondering HOW you controlled for demographic and institutional factors because I didn't see any process ruling them out. Could you include a short explanation about how you did that?

- **Provide constructive feedback on the interpretations and initial conclusion. What is most effective in the presentation of the results? What additional detail can the team provide to make the results and conclusions easier for the reader to understand?**

You did not show the intercept and coefficients for your final model, so it would be helpful if you listed the most significant ones and described how they impact exam scores the most. You say that "x,y,z have measurable effects," yet the reader wants to know which of them have the most effect.

- **What aspect of this project are you most interested in and think would be interesting to highlight in the written report?**

I would love to see the relationship between stress level and exam performance. What is the exact co-efficient, and how does stress level correlate with the mental health category in your EDA?

- **Provide constructive feedback on any issues with file and/or code organization.**

Your eda is beautiful. Would love to see the results of the linear model code.

- **(Optional) Any further comments or feedback?**

This is great!